

Pore characteristics and hydraulic properties of a sandy loam supplied for a century with either animal manure or mineral fertilizers

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Abstract. Application of organic residues to soil is generally assumed to improve soil tilth. Only few studies have reported the long-term effects on the more subtle aspects of soil porosity, and no reports have considered the potential effects of organic amendments on the pore system in the subsoil. We sampled undisturbed soil cores (100 cm³ and 6280 cm³) using metal cylinders in differently fertilized plots in the long-term field experiment at Askov Experimental Station, Denmark. We selected the 0–60 cm soil layer of plots dressed for a century with either mineral fertilizers (labelled NPK) or animal manure (labelled AM) and unfertilized plots (UNF) as a reference. Both fertilization treatments were studied at two levels of nutrient application: ‘normal’ (labelled ‘1’) and 1.5 times ‘normal’ (labelled ‘1½’). Water retention, air permeability and air diffusivity were measured on the small cores, and we used the large cores for measuring near-saturated and saturated hydraulic conductivity. In the plough layer, the AM and NPK soils displayed identical pore volumes in size fractions that were larger as well as smaller than 30 µm, while the UNF soil had a significantly smaller volume of pores < 30 µm. No clear trends were found in treatment effects on pore organization as calculated from air diffusivity and air permeability measurements. No significant differences in hydraulic conductivity were found in the plough layer. For the subsoil below ploughing depth, significantly larger macropore volumes and near-saturated hydraulic conductivities were found for soil of plots receiving the larger (‘1½’) amount of nutrients compared with the ‘normally’ dressed soil. This effect was independent of fertilization system (AM or NPK). We attribute the larger volume of macropores to the improved root growth conditions in the soil with the higher nutrient level. We conclude that addition of animal manure at rates realistic in agriculture has only a modest effect on soil pore characteristics of the plough layer soil compared with the use of mineral fertilizers. For the subsoil below ploughing depth, a high level of nutrient application may increase soil macroporosity and near-saturated hydraulic conductivity, but the origin of nutrients is of no significance.

Keywords: Fertilization, manure, porosity, macropores, tortuosity, hydraulic conductivity

INTRODUCTION

It is generally assumed that the organic manures improve soil quality for farming purposes compared with mineral fertilizers. Organic inputs to soil have been reported to affect soil biota: earthworms (e.g. Christensen *et al.* 1987; Whalen *et al.* 1998), meso-fauna (e.g. Axelsen & Thorup-Kristensen 2000), and microbes (e.g. Martens *et al.* 1992; Goyal *et al.* 1993; Schjønning *et al.* 2002a). Organic matter *per se* plays a dominant role in soil aggregation (Tisdall & Oades 1982), and the level of organic matter in soil has been shown to affect key tilth characteristics such as soil friability (Watts & Dexter 1998).

Several investigators have studied the effects of farmyard manure on soil structure and hydraulic properties (e.g. Mathers & Stewart 1984; Campbell *et al.* 1986; Anderson *et al.* 1990). Some found an increased saturated hydraulic conductivity (Mathers & Stewart 1984; Anderson *et al.* 1990), while others found no effect on hydraulic properties (Campbell *et al.* 1986). For Danish conditions, Schjønning *et al.* (1994) found a tendency for a larger volume of pores < 30 µm in soils receiving animal manure than in soils applied with inorganic fertilizers for more than 90 years. Schjønning *et al.* (2002b) found marked differences in soil pore characteristics of the A-horizon of an organically managed dairy farm soil compared with a neighbouring soil which was cropped continuously with small grain cereals and received no organic manures. However, the study did not allow a differentiation between effects of the crop sequence and the application of manure.

A number of studies reported significant densification of the soil layer just below the ploughing depth (e.g.

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Håkansson *et al.* 1996; Djurhuus & Olesen 2000). The existence of a dense plough-pan may influence air exchange (e.g. Schjønning 1989) and water movement (Comia *et al.* 1994; Ball *et al.* 1998). Munkholm (2000), and Munkholm *et al.* (2001a) demonstrated that a plough-pan might force the crop roots to pass through the compacted layer only via persistent earthworm channels. Similarly, the high density of the pan may cause the preferential flow of percolating water through such vertical macropores (Edwards *et al.* 1992; Petersen *et al.* 2004). An increased biotic activity (including root growth) due to repeated applications of organic residues may be hypothesized to affect also the porosity of the soil layers below tillage depth. However, to our knowledge, there are no published studies on the effects of organic residues on subsoil pore characteristics.

The present investigation examined soil porosity characteristics for soils with different fertilization histories applied for about a century. The aim was to quantify the potential effects of animal manure and mineral fertilizer on soil pore characteristics and the hydraulic properties of the plough layer and the subsoil.

MATERIALS AND METHODS

Soils and treatments

Askov Experimental Station is located in southern Jutland (55°28'N, 09°07'E). The Askov soil is a loamy sand: according to Nielsen & Møberg (1984), the geomorphology classifies as terminal morainial deposits from the Weichselian glacial stage, whereas Sunberg *et al.* describe it as morainial deposits from the earlier saaliun glaciation. The plough layer (0–22 cm) contains about 11% clay (<2 µm), 13% silt (2–20 µm), and 74% sand (>20 µm) of which about half is coarse sand >200 µm. The clay content increases with depth to c. 17% at 60–80 cm, which is attributed to lessivage.

A long-term experiment on animal manure and mineral fertilizers was initiated in 1894. The experiment includes unfertilized plots (UNF) and plots dressed with different amounts of mineral fertilizers (NPK) or animal manure (AM) applied at different rates ($\frac{1}{2}$, 1, $1\frac{1}{2}$, 2). The amounts of nutrients added were changed in 1907, 1923, 1949, and 1973, but within each period the amounts of N, P and K applied to the AM and NPK plots were almost equivalent. Ever since 1973 the term '1 fertilizer' (animal manure or mineral fertilizer) has referred to a total application of about 400 kg N ha⁻¹, 80 kg P ha⁻¹ and 320 kg K ha⁻¹ in a four-year rotation of winter cereals, root crops, spring cereals undersown with a grass-clover mixture, and grass-clover. It should be noted that this level of applied nutrition is less than the rates officially recommended for achieving optimum yields (Christensen *et al.* 1994). From 1973, farmyard manure was replaced by animal slurry. The four crops in the rotation are grown every year. Hence, the fertilization treatments are applied in each of four fields (labelled B2, B3, B4 and B5). The present study included all four fields, and we chose to study AM and NPK plots at nutrient levels '1' and '1½', and also UNF plots.

Detailed accounts of the Askov long-term field experiments have been provided by Christensen *et al.* (1994) and

Christensen (1997); the former also reviewed previously published results on crop yields. Schjønning *et al.* (1994) and Munkholm *et al.* (2002) described soil mechanical properties measured in fields B2, B4 and B5.

Sampling

Sampling took place in each of the years 1995–98 and 2002. Soil cores (100 cm³, inner diameter 6.10 cm, depth 3.42 cm) were taken at 8–12 cm depth in metal cylinders hammered into the soil. The cylinders were held in position by a special flange to ensure only vertical movement within the soil. After careful removal, the end surfaces were trimmed with a knife.

In 2002, sampling of cores also took place at the 28–32 cm and 48–52 cm depths; at the same time, larger cores (6280 cm³, inner diameter 20 cm, depth 20 cm) were collected for the 0–20, 20–40 and 40–60 cm depths. The larger sampling cylinders were forced into the soil with a hydraulic press mounted on a tractor. All samples were protected from evaporation and physical disruption and stored at 2–5 °C until analysis. The 2002 sampling included collection of bulk soil from the 20–40 and 40–60 cm layers. Table 1 gives an overview of the sampling programme.

Field measurements

On 11 May 1998, water infiltration was measured with a tension infiltrometer at each of the chosen plots in field B2 (Table 1), when the barley plants were at the early tillering stage. The tension infiltrometer was placed between rows on a thin layer of moist sand to ensure good hydraulic contact. Steady-state flow rates from the infiltrometer were then measured at three matric water potentials: +1 (saturation), –3, and –6 hPa. Calculation of the hydraulic conductivities was based on the method of Ankeny *et al.* (1991).

Laboratory measurements

Bulk soil from the 20–40 and 40–60 cm layers sampled in 2002 was air-dried and passed through a 2 mm sieve. Total carbon (C) content was determined on ball-milled subsamples using a LECO CNS-1000 analyzer with IR detector (LECO Corporation, St Joseph, MI). Each sample was tested for carbonate-C content.

The undisturbed 100-cm³ core samples were saturated on tension tables with capillary water from beneath. The cores were then drained in steps to the matric potentials listed in Table 1. Tension tables and ceramic plates were used for matric potentials exceeding and less than –100 hPa, respectively. The soil cores were finally oven-dried at 105 °C for 24 hours. The weights of the samples were recorded at each matric potential and after oven-drying. Soil porosity was estimated from actual bulk density and particle density measured according to Schjønning *et al.* (1994). The volumetric water content at each matric potential was taken as the respective difference in weight with oven-dried samples. Air-filled porosity, ϵ_a , at a given matric potential was calculated as the difference between porosity and the water retained at the specific potential.

After equilibration at –100 hPa matric potential, air permeability (K_a) was measured by the steady-state method of

Table 1. Overview of sampling and measurement programme.

Sampling date	Field label	Number of replicates ^a	Crop grown	Treatments examined ^b	Soil layer sampled (cm)	Water retention at matric potential (- hPa)	Air permeability measured at potential (- hPa)	Gas diffusivity measured at potential (- hPa)	Hydraulic conductivity measured	Field measurement with tension infiltrometer
10 April 1995	B3	3 (3)	Winter wheat established autumn 1994	UNF, NPK1½, AM1½	8–12	100	100			
7 May 1996	B5	3 ^c (3)	Spring barley sown in April; ^f soil ploughed in autumn 1995	UNF, NPK1½, AM1½	8–12	100				
7 April 1997	B4	3 ^d (6)	Winter wheat established autumn 1996	UNF, NPK1½, AM1½	8–12	10, 30, 50, 100, 300, 1000	10, 30, 100, 300	10, 30, 100, 300		
11 May 1998	B2	3 (6)	Spring barley sown in April; ^f soil ploughed in autumn 1997	UNF, NPK1½, AM1½	8–12	10, 30, 50, 100, 300, 1000	10, 30, 100, 300	10, 30, 100, 300		X
25 November 2002	B5	3 ^e (6)	Winter wheat stubble	UNF, NPK1 NPK1½, AM1 AM1½	8–12 (0–20) ^g 28–32 (20–40) 48–52 (40–60)	10, 30, 50, 100, 300, 1000	100		X	

^aNumber of replicate plots per treatment with number of replicate 100 cm³ cores within each plot in parentheses; only one 6280 cm³ core (for hydraulic conductivity) was sampled for each combination of plot and soil layer. ^bUNF = unfertilized; NPK = dressed with mineral fertilizer; AM = dressed with animal manure; '1' = normal nutrient level; '1½' = 50% extra nutrient. ^cOnly partly correct: an NPK1 plot was used as the third 'NPK1½' plot. ^dOnly partly correct: an AM1 plot was used as the third 'AM1½' plot. ^eAM1½ only 2 replicate plots. ^fSampling below tillage depth. ^gInterval in parentheses refers to large cores.

Iversen *et al.* (2001). Prior to measurement, the soil was gently pressed to the edge of the metal ring to minimize the risk of air leaking through the ring (Ball & Schjønning 2002). The cores sampled in 1997 and 1998 were analysed for air permeability and gas diffusivity at -10, -30, -100 and -300 hPa. For the latter analysis, we applied the non-steady state method suggested by Taylor (1949), using equipment described by Schjønning (1985a) and oxygen as the diffusing gas. Prior to measurement, we flushed the chamber above the soil core with oxygen-free nitrogen. Our calculations for similar, undisturbed soil cores indicated that we could ignore potential error due to oxygen consumption in the cores during the diffusion.

The specific air permeability (Groenevelt *et al.* 1984), also named 'pore organization' ($PO = K_a/\epsilon_a$; Blackwell *et al.* 1990), and 'specific gas diffusivity' ($[D_a/D_0]/\epsilon_a$; Gradwell 1961) was calculated from the permeability and diffusivity measurements. The relative diffusivity, D_a/D_0 , is the quotient of the diffusion coefficient in soil, D_a , and that in air, D_0 .

The larger, undisturbed 6280-cm³ soil cores were slowly saturated in a drainage box with a ceramic plate over a period of 3 days. Then the cores were drained to a matric potential of -20 hPa and removed to the drip infiltrometer where the cores were placed on a ceramic plate. The drip infiltrometer (van den Elsen *et al.* 1999) measured the unsaturated hydraulic conductivity (k) in the wet (near saturated) range at matric potentials (h) between approximately -10 and -100 hPa. Five tensiometers were placed in the soil core and a water flux applied with a needle device to the top of the sample. The measurement was started as close to saturation as possible while applying a constant suction to the bottom of the soil core. When a steady-state pressure head had been reached, the flux density was determined. This, together with measurements of the pressure head gradient, yielded the hydraulic conductivity at the resulting matric potential. Subsequently, the flux was decreased and the procedure described above repeated measuring at a lower pressure head (see Iversen *et al.* 2004 for details).

After the measurement of k , the soil cores were placed in the device used for the measurement of the saturated hydraulic conductivity (K_s). The soil cores were re-saturated overnight and K_s determined using the constant head method of Klute & Dirksen (1986). For further details consult Iversen *et al.* (2004).

Statistical analyses

We used results of previous studies with a large number of replicate samples to decide whether our data should be transformed before analysing them statistically. Based on this knowledge, the air permeability, the pore organization ($PO = K_a/\epsilon_a$) and the saturated hydraulic conductivity were log-transformed. Averages were calculated for each of the plots and used for tests of management effects in a one-way analysis of variance (procedure 'GLM' in SAS 1996).

Statistical tests on the measurements of the unsaturated hydraulic conductivity, k , on the large soil cores were carried out in two steps. First, a likelihood ratio test on the

linearity between h and $\log(k)$ was performed by means of a test for equality of an exponential curve model and a model with a free curve of response for h . Differences in the slope and the interception in relation to management effects were tested using an exponential curve model with block effect. Subsequently, differences between predicted values of k caused by the interaction between the two fertilization systems and nutrient levels were found for the individual depths. All tests were performed using the procedure 'MIXED' in SAS (1996).

RESULTS

Soil content of organic C

Table 2 gives soil content of organic C for selected plots in field B5. For the plough layer soil (data from Schjøning *et al.* 1994), a significantly smaller content of organic C was found in the UNF soil compared with the AM or NPK soils. Within each fertilization system, increased application of nutrients ('1' vs '1½') has increased the content of organic C. This trend was not found for the soil layers below ploughing depth. For plough layer soil, as well as for the two subsoil layers, the AM treatment gave higher concentrations of organic C than the NPK treatment. Although not statistically significant, the relative AM-induced elevation of organic C was most pronounced for the two subsoil layers. For the plough layer, the AM soil had ~10% more organic C than the NPK soil. For the two subsoils, the difference was ~35%.

Pore size distribution of plough layer soil

None of the samplings displayed any effect of fertilization on the volume of macropores $>30\ \mu\text{m}$ (Table 3). The smallest volume of pores $<30\ \mu\text{m}$ (water held at a matric potential of $-100\ \text{hPa}$) was found for the unfertilized (UNF) treatment at all four fields in 1995–98. This effect was significant for the B3 and B4 fields and when averaged for all four fields (Table 3). On an average basis, there was no significant difference between the NPK and AM treatments. In accordance with the above, the total porosity was significantly higher for the fertilized treatments (NPK & AM) than for the unfertilized soil (UNF).

The measurements in 1997, 1998 and 2002 allow for a more detailed analysis of the pore size distribution. Figure 1 shows that the volume of macropores $>30\ \mu\text{m}$ was much

higher at the 1997 and 1998 samples than in 2002. This can be ascribed to the sampling date (Table 1): the B4 and B2 fields were sampled in the spring (1997 and 1998, respectively) some 4–6 months after ploughing in the autumn, while the B5 field (2002) was sampled in the autumn, more than a year after ploughing. Obviously, the volume of macropores had been consistently reduced during the summer period. As sampling deliberately avoided visible tractor tracks, the effect may be ascribed to natural settling.

In accordance with Table 3, no treatment effects on the volume of macropores were statistically significant in the more detailed study shown in Figure 1, but some effects of fertilization on the smaller pores appeared. However, in 1998 the effects seemed to neutralize each other, as there was no significant effect on the sum of pore volume from the two size classes ($3\text{--}30\ \mu\text{m}$). In 1997, the largest volume for both these size fractions was found with the AM1½ treatment, whereas in 2002 this was the case for the NPK1½ treatment.

Pore tortuosity and organization of plough layer soil

All samplings allow for the calculation of the specific air permeability at a matric potential of $-100\ \text{hPa}$ ($PO_{100} = K_{a,100}/\varepsilon_{a,100}$). A high value of PO_{100} indicates that the macropores $>30\ \mu\text{m}$ are effective in conducting air by convection. This may in turn be interpreted as a high degree of continuity (organization) of the pores. From Table 3 it appears that neither the air permeability *per se* nor the pore organization of plough layer soil was significantly ($P = 5\%$) affected by the fertilization treatments. However, generally there was a trend towards a high pore organization for the AM treated plots. Averaged for the 1995–98 samplings, this trend was significant ($P = 5.2\%$; Table 3).

The specific gas diffusivity ($[D_a/D_0]/\varepsilon_a$) was suggested by Gradwell (1961) as an expression of pore tortuosity. A high value of the specific gas diffusivity implies a non-tortuous (linear) course of the pores conducting the gas diffusion. In 1997, there was a general trend towards higher specific diffusivity for the fertilized treatments (NPK & AM) than for the unfertilized soil (UNF) (Figure 2). The effect was significant for the -100 and $-300\ \text{hPa}$ potentials (right-hand data points). The NPK and AM treatments could not be significantly distinguished. The AM treatment also displayed the highest index in 1998 (field B2) and could be significantly differentiated from the NPK and UNF treatments at the -30 and $-100\ \text{hPa}$ potentials. At the $-300\ \text{hPa}$ potential (far-right data points), the AM and UNF soils displayed significantly higher specific diffusivities than the NPK soil.

The pore organization expressed as the specific air permeability ($PO = K_a/\varepsilon_a$) at the four potentials investigated in 1997 and 1998 displayed the same trend as the specific gas diffusivity (Figure 2). For both years, the AM treatment generally had the highest pore organization, while the UNF – and in 1998 the NPK – treatment had lower values. However, the only statistically significant effect appeared in 1998 at a water potential of $-30\ \text{hPa}$, where AM treatment had higher values than the NPK and UNF treatments.

Table 2. Organic carbon ($\text{g C } 100\ \text{g}^{-1}$ soil) for different fertilization treatments in field B5.

Depth (cm)	Treatment ^a				
	AM1	AM1½	NPK1	NPK1½	UNF
0–20 ^b	1.30cd	1.35d	1.15b	1.25c	1.07a
20–40 ^c	0.57a	0.56a	0.45a	0.39a	0.44 ^e
40–60 ^d	0.25a	0.25a	0.20a	0.17a	n.d.

^aFor key to treatments, see text and Table 1. ^bSamplings from 1985 (Schjøning *et al.* 1994). ^cAverage of samplings from 1985 and 2002.

^dSamplings from 2002. ^eFrom 1985 sampling only; cannot be statistically compared with the four other treatments. Figures followed by the same letter are not significantly different ($P = 5\%$). n.d. = not determined.

Table 3. Soil pore volume in two size fractions, and air permeability, $K_{a,100}$, and pore organization, $PO = K_{a,100}/\epsilon_{a,100}$, at -100 hPa, where $\epsilon_{a,100}$ is air-filled pore space at that potential. Soil cores were sampled in the plough layer in 100-cm^3 steel cylinders. Air permeability and pore organization are given as geometric means (log-normal distribution). See Table 1 for number of replicates.

Year	Field	Treatment ^a	Pore volume ($\text{m}^3 \text{m}^{-3}$)			Air permeability (μm^2)	Pore organization (μm^2)
			Pores $< 30 \mu\text{m}$	Pores $> 30 \mu\text{m}$	Porosity		
1995	B3	AM1½	0.290a	0.083a	0.373a	12.8a	167a
		NPK1½	0.321b	0.079a	0.399a	4.1a	57a
		UNF	0.273a	0.095a	0.368a	2.9a	34a
1996	B5	AM1½	0.244a	0.197a	0.441a	17.4a	93a
		NPK1½ ^b	0.236a	0.189a	0.425a	14.8a	79a
		UNF	0.226a	0.174a	0.400a	13.8a	81a
1997	B4	AM1½ ^c	0.289a	0.159a	0.448a	28.4a	180a
		NPK1½	0.270b	0.159a	0.429ab	33.2a	211a
		UNF	0.253c	0.166a	0.419b	18.3a	113a
1998	B2	AM1½	0.286a	0.166a	0.452a	6.9a	43a
		NPK1½	0.263a	0.184a	0.448a	3.7a	21a
		UNF	0.256a	0.159a	0.416a	2.7a	17a
Average: 1995–98	Average: B2–B5	AM1½	0.277a	0.151a	0.428a	14.4a	105a
		NPK1½	0.272a	0.153a	0.425a	9.3a	67b
		UNF	0.252b	0.149a	0.401b	6.7a	48b
2002	B5	AM1	0.288a	0.102a	0.390a	7.4a	75a
		AM1½	0.293a	0.095a	0.388a	3.8a	41a
		NPK1	0.276a	0.095a	0.370a	3.8a	42a
		NPK1½	0.287a	0.102a	0.389a	3.6a	37a

^aFor key to treatments, see text and Table 1. ^bTwo replicate plots with NPK1½ and one with NPK1. ^cTwo replicate plots with AM1½ and one with AM1. Figures with the same letters within each column and year are not statistically different ($P = 5\%$).

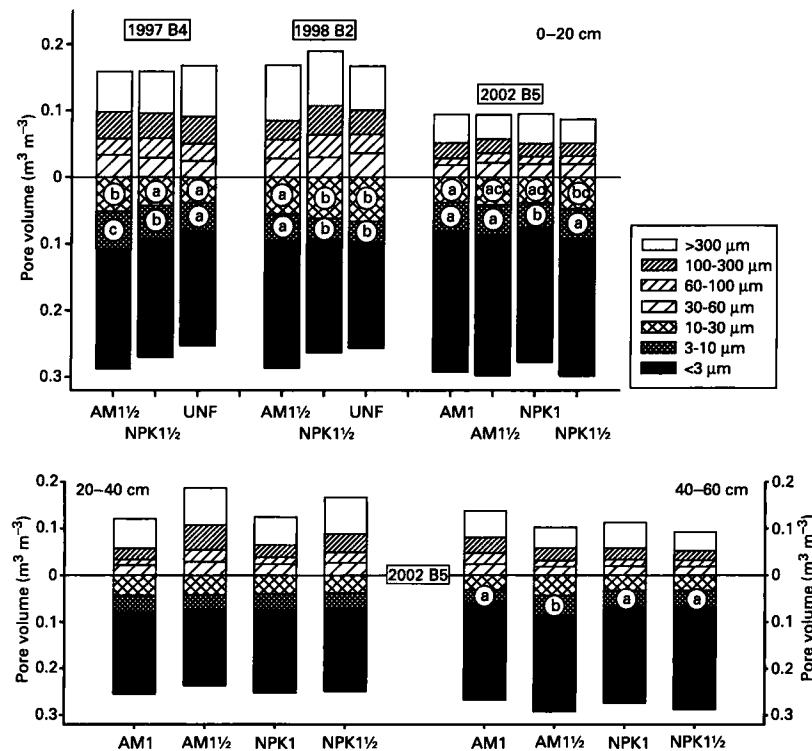


Figure 1. Pore size distribution for the differently fertilized soils. Volumes of size fractions within the same year shown with the same letter are not statistically different ($P = 5\%$). Absence of lettering in a size fraction means that no significant effect could be detected for that size class. UNF = unfertilized; NPK = dressed with mineral fertilizer; AM = dressed with animal manure; '1' = normal nutrition level (consult text); '1½' = 50% extra nutrients.

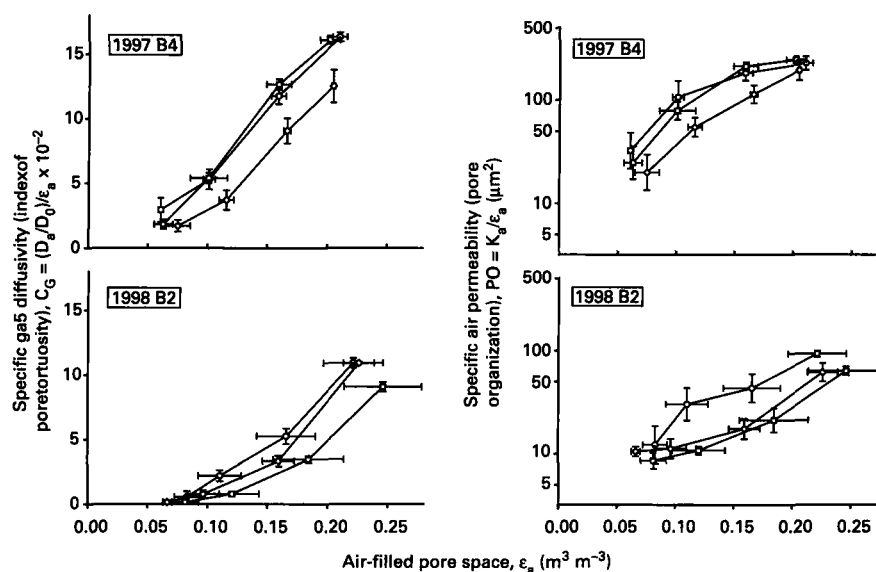


Figure 2. Specific gas diffusivity ($[D_a/D_0]/\epsilon_a$) and specific air permeability ($PO = K_a/\epsilon_a$; geometric means) at four matric potentials (data points from left to right: -10 , -30 , -100 , -300 hPa) for the different fertilization treatments investigated in fields B4 and B2. Bars indicate standard error of the mean (calculated on log-transformed data for the specific gas permeability) across replicate plots ($n = 3$). Treatments are: $\circ$, AM1½; $\square$, NPK1½; $\diamond$, UNF (for key, see text and Table 1).

Water infiltration in the field

No significant effect of the fertilization treatments could be detected in the hydraulic conductivity of the surface soil at suctions close to saturation (Table 4). The UNF soil displayed the *highest* conductivity when the infiltration took place at matric potentials of -3 and -6 hPa. The AM and NPK treatments could not be significantly distinguished.

Pore size distribution and pore organization of soil below ploughing depth

No significant effect of fertilization treatments could be detected for the 20–40 cm layer (Figure 1). In the 40–60 cm layer, the AM1½ treatment displayed a larger volume of 3–10 μm pores than the three other treatments investigated.

Although not significant when analysing the four treatments as parallel effects, Figure 1 indicates higher volumes of macropores for the AM1½ and NPK1½ treatments in the 20–40 cm layer compared with the two treatments receiving only level '1' addition of nutrients. Table 5 shows the result of an alternative analysis of variance with a model including the effect of 'system' (AM or NPK) as well as 'level of nutrition' ('1' or '1½'). It appears that no

'system' effect could be detected, while the probabilities (P -values) of higher volumes of macropores for the '1½' compared to the '1' level are close to the $P = 5\%$ level for most fractions of the 20–40 cm layer. The $P < 5\%$ criteria of significance was met for the sum of pores $> 30 \mu\text{m}$ (Table 5). Despite the differences in volumes of macropores, no significant effects of either the air permeability or the pore organization (at -100 hPa potential) could be found (Table 6).

Hydraulic conductivity of large-scale soil cores

Measured unsaturated hydraulic conductivities (k) in field B5 at the three depths examined are shown in Figure 3. Statistical tests showed that a linear relation existed

Table 4. Hydraulic conductivity (cm d^{-1}) at three matric potentials as measured with a tension infiltrometer in the surface soil of field B2 (1998).

Treatment ^a	Matric water potential (hPa)		
	+1	-3	-6
AM1½	272a	8.82a	3.32ab
NPK1½	266a	6.65a	0.85b
UNF	281a	19.60b	7.46a

^aFor key to treatments, see text and Table 1. Figures with the same lettering within each matric potential are not statistically different ($P = 5\%$).

Table 5. Soil pore volume in different size fractions $> 30 \mu\text{m}$ as influenced by fertilization system and level of fertilization. Soil cores sampled in 100 cm^3 cylinders in the 20–40 cm layer (28–32 cm depth) and 40–60 cm layer (48–52 cm depth) of field B5 (2002).

Depth (cm)	Pore size fraction (μm)	Pore volume ($\text{m}^3 \text{m}^{-3}$)					
		Fertilization system ^a			Nutrient level ^a		
		AM	NPK	P (%)	'1'	'1½'	P (%)
20–40	> 300	0.071	0.069	> 15	0.061	0.079	5.4
	300–100	0.037	0.033	> 15	0.025	0.046	7.2
	100–60	0.019	0.019	> 15	0.013	0.024	5.4
	60–30	0.025	0.026	> 15	0.023	0.028	> 15
	> 30 (sum)	0.152	0.147	> 15	0.122	0.177	4.2
40–60	> 300	0.050	0.048	> 15	0.056	0.042	> 15
	300–100	0.030	0.022	> 15	0.028	0.023	> 15
	100–60	0.019	0.013	> 15	0.019	0.013	> 15
	60–30	0.021	0.019	> 15	0.021	0.019	> 15
	> 30 (sum)	0.120	0.102	> 15	0.125	0.097	> 15

^aFor key to treatments, see text and Table 1. No interactions between fertilization system and nutrient level were significantly different ($P = 5\%$).

Table 6. Air permeability at a matric potential of -100 hPa, $K_{a,100}$, and pore organization, $PO = K_{a,100}/\epsilon_{a,100}$, where $\epsilon_{a,100}$ is air-filled pore space at that potential. Both parameters are given as geometric means (log-normal distribution). Soil cores sampled as per Table 5.

Soil layer (cm)	Treatment ^a	Air permeability (μm^2)	Pore organization (μm^2)
20–40	AM1	18.4	149
	AM1½	26.0	147
	NPK1	15.6	117
	NPK1½	19.4	117
40–60	AM1	19.5	156
	AM1½	11.0	109
	NPK1	11.2	98
	NPK1½	8.2	81

^aFor key to treatments see text and Table 1. No treatment effects are significant ($P = 5\%$).

between $|h|$ and $\log(k)$ at all three depths. No treatment effects were found for the plough layer (0–20 cm). For the 20–40 cm depth, an effect of treatment was found on slope for all combinations of treatments ($P = 5\%$). For this depth, the interception (predicted hydraulic conductivity at suction = 0) was found to be significantly higher for the '1½' than for the '1' level of nutrition ($P < 0.1\%$), while no effect of fertilization system (AM vs NPK) could be detected. At the 40–60 cm depth, significant slope differences were found for all combinations of treatments except between NPK1 and NPK1½. Also for this depth, the interception was found to be higher for the '1½' than for the '1' level of nutrition but the effect was not significant ($P = 7.5\%$). As with the 20–40 cm depth, no effect of fertilization system (AM vs NPK) could be detected.

Table 7 compares predicted values of $k(h)$ between the two fertilization systems (AM vs NPK) and between the two different nutrient levels ('1' and '1½'). Values are shown for three different levels of $|h|$. Since no treatment effects were found for the 0–20 cm depth, results are only shown for the two lower depths.

Generally, no significant interactions were found between the fertilization system and the nutrition level, that is, the trend of nutrition level effect was the same in both fertilization systems. At the 20–40 cm depth, k was predicted to be significantly greater for the '1½' than the '1' level of nutrient input and at all three water suctions (Table 7). Also for the 40–60 cm depth, k was always predicted to be largest for the high level of nutrient application. However, at this depth the effect was only significant at the dry end of the investigated range of matric water potentials. The level of k at this depth and these water potentials was very small (Table 7), and so the increase due to elevated nutrition may have important impacts. The test of differences between fertilization systems (NPK and AM) generally showed no significant effects except for the 40–60 cm depth at $h = -20$ hPa.

The results from the measurements of saturated hydraulic conductivity displayed interactions between fertilization system and level of nutrition for the 0–20 and 40–60 cm depths. Although this interaction was statistically significant only for the 40–60 cm depth (analysis not shown), we analysed the effect of nutrition level separately for each

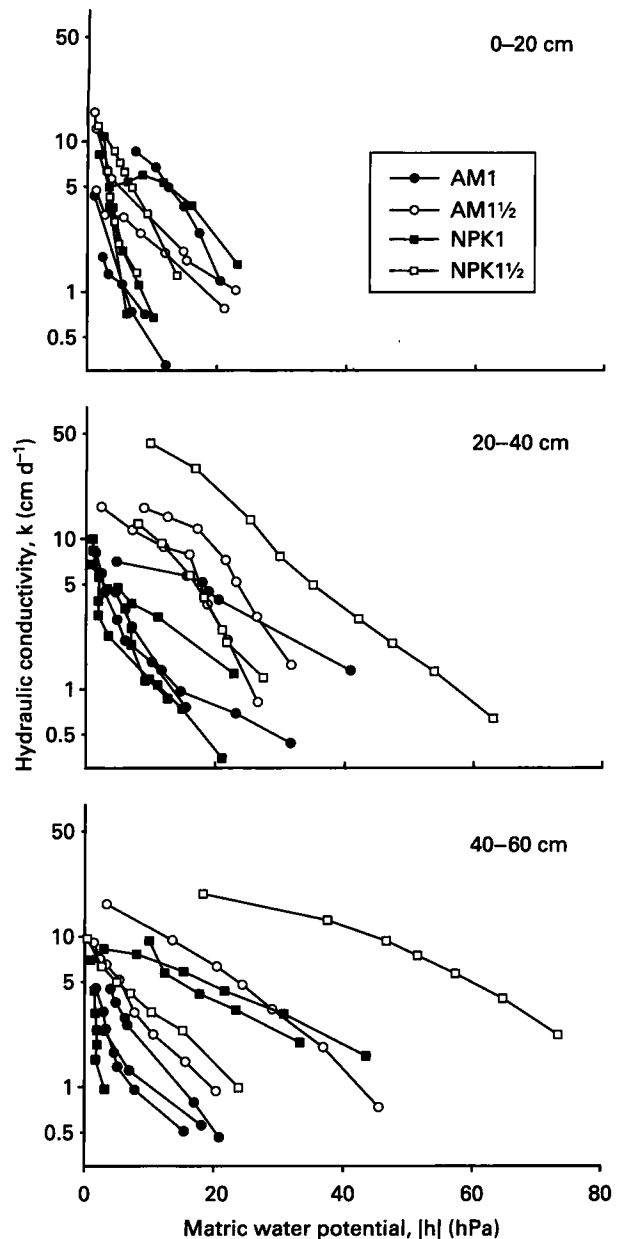


Figure 3. Unsaturated hydraulic conductivity, k , measured for a range of matric water potentials, h , in field B5 (2002). Each line represents one large soil core. For key to treatments, see text and Table 1.

fertilization system (Table 8). For four of the six combinations of soil depth and fertilization system, the application of 50% extra nutrients compared with the normal level '1' increased the saturated hydraulic conductivity. However, the trend was only significant for the AM system at 40–60 cm depth (Table 8).

DISCUSSION

Plough layer soil

Animal manure had no effect on the volume of macropores $> 30 \mu\text{m}$. Even when comparing manured plots with those

Table 7. Predicted unsaturated hydraulic conductivity, k (cm d^{-1}) for different values of the soil matric water potential $|h|$, using a model with linear relation between $\log(k)$ and h . Soil cores were sampled in 6280 cm^3 cylinders in the 20–40 cm and the 40–60 cm layer of field B5 (2002).

Depth (cm)	$ h $ (hPa)	Fertilization system ^a			Nutrient level ^a		
		AM	NPK	P (%)	'1'	'1½'	P (%)
20–40	0	16.5	16.4	> 15	6.3	42.8	< 0.1
	10	6.6	6.1	> 15	2.5	16.0	< 0.1
	20	2.6	2.2	> 15	1.0	6.0	< 0.1
40–60	0	8.1	10.9	> 15	5.2	17.0	7.5
	10	2.9	7.0	> 15	2.2	9.0	3.7
	20	1.0	4.5	2.6	1.0	4.7	1.9

^aFor key to treatments, see text and Table 1. No interactions between fertilization system and nutrient level were significant ($P = 5\%$).

not receiving any fertilizers for a century, the volume of macropores was not significantly affected (Table 3). The result is in accordance with the samplings in field B5 reported by Schjønning *et al.* (1994). Previous measurements indicated significantly higher wet aggregate stability for the AM than the NPK treated soil (Schjønning 1995). However, the present results show that the sandy loam at Askov has a very high natural density of $\sim 1.66 \text{ g cm}^{-3}$ (estimated from porosity data in Table 3) when left undisturbed for only a year. This indication of generally poor structural stability may explain why the fertilization effects are not manifested in the volume of macropores. Franken & Hurtmanns (1985) found a tendency for a higher volume of macropores $> 50 \mu\text{m}$ in plots receiving animal manure compared with plots dressed only with mineral fertilizer. However, the effect was not found in all of the sampling events they reported. Miller *et al.* (2002) reported an increased volume of $> 1120 \mu\text{m}$ pores but only at rather high rates of manure application. Based on these results, manure effects on soil macroporosity seem to be rather weak.

Our data show a clear decrease in water content at the -100 hPa potential (pores $< 30 \mu\text{m}$) for the UNF treatment compared with the AM and NPK treatments (Table 3). No significant difference could be detected between the AM and NPK treatments. The data from field B4 in 1997 (Figure 1) indicate a significantly higher volume of $3\text{--}30 \mu\text{m}$ pores for the AM treatment than for the NPK treatment. This is in accordance with results from field B5 reported by Schjønning *et al.* (1994), who found a tendency for a higher volume of $0.2\text{--}30 \mu\text{m}$ pores in AM than in NPK plots. Celik *et al.* (2004) found that compost and

manure additions to a clay soil increased the water retained in the pore size fraction $0.2\text{--}10 \mu\text{m}$. Based on these results, the amount of water available to plants seems to be slightly increased when adding organic manures instead of mineral fertilizers.

The data from the former study at field B5 (Schjønning *et al.* 1994) included a measurement of the water retained at the -1.5 MPa potential (the 'wilting point'). It appeared that the difference in water-holding capacity at -100 hPa could be ascribed mainly to differences already present at -1.5 MPa . A number of studies have indicated that organic matter additions to soil increase the water retained at this potential (e.g. Jacobowitz & Steenhuis 1984; Bhatnagar *et al.* 1992). This means that organic matter additions primarily induce higher water contents at the wilting percentage ($< 0.2 \mu\text{m}$ pores) and not in the water content ($0.2\text{--}30 \mu\text{m}$ pores) available to plants (Miller *et al.* 2002). Metzger & Yaron (1987) arrived at the same conclusion based on a review including a range of studies with different types and amounts of organic residues.

Our investigations on pore characteristics indicated a tendency for reduced continuity and/or increased tortuosity of the unfertilized soil compared with the NPK and AM treatments (Table 3, Figure 2). Mostly, the AM treatment was superior to the NPK treatment but the effect was seldom statistically significant. The literature on pore characteristics in relation to soil organic matter is scarce. Schjønning (1985b, 1986) reported a tendency towards higher permeability and continuity of pore systems in soil where straw had been incorporated for a number of years compared with the removal of straw. No effects on pore characteristics could be measured from 18 years of repeated incorporation of a catch crop of Italian ryegrass (*Lolium multiflorum*; Schjønning 1989).

Some papers report increased saturated hydraulic conductivities for soil that has received organic residues (e.g. Shirani *et al.* 2002; Celik *et al.* 2004), but the reports provide questionable results because of the high variability inherent in the determination of K_s (Metzger & Yaron 1987). Furthermore, most studies have addressed the effects of very high rates of organic dressings (Khaleel *et al.* 1981). The results from the present study indicate that the bulk plough layer soil ($8\text{--}12 \text{ cm}$ depth) is not affected by animal manure applied at rates realistic for repeated additions in normal agricultural practice (Figure 3 & Table 8). This observation was reached when comparing soil dressed with equivalent amounts of mineral fertilizers. It seems that large rates of organic residues are needed to induce statistically significant changes in K_s (Chang *et al.* 1983; Jacobowitz & Steenhuis 1984; Wei *et al.* 1985; Miller *et al.* 2002).

Our results on infiltration of water into the surface soil are difficult to interpret. We found a significantly greater infiltration capacity for the UNF treatment compared with the NPK and AM treatments at the -3 and -6 hPa tensions (Table 4). This may indicate differences in crust formation on the surface, but we do not have evidence of this and crusting would be more likely for the UNF soil due to its lower content of organic C. Although the barley plants were juvenile at the time of measurement, a possible cause may be the blocking of macropores by supposedly

Table 8. The effect of nutrient level on saturated hydraulic conductivity (cm d^{-1}) for each fertilization system analysed separately. Data are geometric means.

Depth (cm)	AM system ^a Nutrient level		P (%)	NPK system ^a Nutrient level		P (%)
	'1'	'1½'		'1'	'1½'	
0–20	3442	1372	> 15	440	8013	11.3
20–40	570	2046	> 15	300	3468	> 15
40–60	99	1949	3.8	1332	186	> 15

^aFor key to treatments, see text and Table 1.

more prolific development of plant roots in the fertilized treatments (NPK and AM). Regardless of this, the results confirm the laboratory measurements on the hydraulic conductivities regarding the (lack of) difference between the NPK and AM treatments. In accordance with this, Miller *et al.* (2002) found no statistically significant effects of animal manure on near-saturated hydraulic conductivity when dressed at recommended rates for a period of 24 years.

Subsoil

It appears that the volume of pores $> 30 \mu\text{m}$ (Table 5) and the near-saturated hydraulic conductivity (Table 7) for the 20–40 cm soil layer are significantly affected by the amount of nutrients applied to the soil. The nutrient level '1½' has given rise to higher conductivities at near-saturated conditions. The same picture was found for the 40–60 cm layer for the hydraulic conductivity, although with weaker statistical significance (Table 7). However, for this soil layer, no significant effect on soil pore volumes was observed (Table 5). Surprisingly, it seemed of minor or no importance whether the nutrients had been applied as animal manure or mineral fertilizers (Tables 5 & 7).

Our measurements do not allow a direct quantification of the number and volume of macropores. However, the data in Table 5 indicate that for the 20–40 cm depth a nutrient-level induced increase in pore volume has taken place for all size classes above $30 \mu\text{m}$; this effect is close to statistical significance, not only for the very large pores $> 300 \mu\text{m}$ but also for pores in the range 60–300 μm (Table 5). In accordance with this, the unsaturated hydraulic conductivity is significantly higher for the '1½' treatments also at matric potentials of -10 to -30 hPa (100–300 μm pores) (Figure 3 & Table 7). This is important for the risk of preferential flow in these soils. The soil having received more nutrients may be better able to conduct water in the matrix of the soil and hence reduce the risk of preferential flow in very large macropores. This would especially become evident if the hydraulic conductivity were low. In this study, we observed $k < 1 \text{ cm d}^{-1}$ at h -values rather close to saturation (Figure 3).

The predicted values of unsaturated hydraulic conductivity in Table 7 are influenced by differences in the intercept (predicted value at $h = 0$) as well as the slopes of the linear model between $\log(k)$ and h . It is noticeable that the slopes of the $k(h)$ plots are distinctly different (Figure 3 and related text in Results). We found a poor correlation between the estimated $k(h)$ slopes and the pore volumes in fractions of pores in the range $> 30 \mu\text{m}$ (analysis not shown). The differences in slope are probably to a great extent influenced by the tortuosity of the soil pores that conduct water at a given matric potential. However, an analysis of this would require measurements of tortuosity characteristics for the large soil cores used for the hydraulic conductivity. In our study, we only had estimates of pore characteristics from the separately sampled 100-cm³ cores.

We interpret the higher volume of macropores for the 20–40 cm layer of those plots that received 50% more nutrients as being due to better conditions for root growth. Well-developed roots may create persistent, large pores in the subsoil. Christensen *et al.* (1994) showed that the yields

of small grain cereals (grain as well as straw) were highest for the NPK rather than for the AM treatment at identical nutrient levels. The opposite was the case for grass–clover yield, while the yield of root crops was similar for the two fertilization systems. All crop yields increased concomitantly with nutrient level. Apparently, the net effect of these expressions of crop development is identical root growth (and subsequently persistent macropores) for the two fertilization systems, but with a significant increase in root growth at greater levels of nutrient application for both systems.

Soil biota are known to thrive at high levels of organic matter. The earthworm population would be expected to increase due to manure application (e.g. Christensen *et al.* 1987; Hansen 1996). Apparently, this increase primarily includes earthworm species living in the plough layer, which is in accordance with Hansen (1996). Deep-burrowing species would probably have produced higher volumes of very large macropores for manured than for mineral fertilized soil. In fact, a *smaller* value of unsaturated hydraulic conductivity was estimated for the AM soil compared with the NPK soil at the 40–60 cm depth (Table 7).

The highest values of unsaturated hydraulic conductivity at 40–60 cm depth were estimated for the high nutrient level (Table 7), despite comparable or even lower volumes of macropores (Table 5). This confirms our speculations above that, in addition to pore volumes, pore tortuosities may also be important for the resulting ability of the soil to conduct water.

General considerations

Our results clearly show that soil not receiving plant nutrients for a century will display different soil quality attributes compared with a soil dressed with nutrients. However, when comparing a soil receiving mineral fertilizers to one receiving animal manure at identical levels of nutrient addition, little variation was found. We have previously investigated the long-term effect of manuring on soil mechanical properties using the same field experiments as reported on here (Schjønning *et al.* 1994; Munkholm *et al.* 2002). Those studies showed that for soil properties such as aggregate tensile strength, friability, bulk soil shear strength and soil plasticity, long-term lack of fertilizer application to soil will also reveal marked differences from soil receiving plant nutrients at rates realistic in agriculture. However, only small differences were detected between soil treated with animal manure and soil dressed with mineral fertilizers.

Edmeades (2003) reviewed the results from 14 long-term trials with application of mineral fertilizers and animal manure. The trials represented a range in soil types although primarily within America and England. Based on his comparisons, no differences in soil productivity and only minor changes in soil quality could be identified. However, Munkholm *et al.* (2001b) and Schjønning *et al.* (2002a, b) found considerable decreases in soil quality attributes in a sandy loam soil cropped continuously with small grain cereals for several decades. This was despite annual applications of mineral fertilizers in dressings equivalent to crop demands. The sustained acceptable soil quality in the NPK-dressed plots in the Askov experiment may be due to a versatile crop

rotation, including a one-year grass–clover ley. Hence, based on our observations, crop rotation aspects should be included in any evaluation of the sustainability of cropping systems where only mineral fertilizers are applied.

Our results showed that the subsoil pore characteristics were generally unaffected by the form of plant nutrients applied, whether organic (animal manure) or inorganic (mineral fertilizers). In contrast, the long-term application of nutrients at a 50% higher level yielded larger volumes of macropores in the subsoil and accordingly increased near-saturated conductivity for water. This may be interpreted as a reduced risk of preferential flow and hence a reduced potential loss of, for example, particle-associated phosphorus and pesticides.

Additional information gleaned from this study is the lower variability in near-saturated than in saturated hydraulic conductivity measurements. Similarly to most of the papers consulted, our results on saturated conductivity gave no clear answer to the questions raised in our study. In contrast, the data for near-saturated conductivity in the form of $k(h)$ gave valuable information. We encourage future studies to include such approaches and ideally also to determine water retention on the cores used for hydraulic conductivity measurements.

CONCLUSIONS

1. Long-term addition of animal manure at realistic rates has only a modest effect on soil-pore characteristics of the plough layer compared with the use of mineral fertilizers.
2. Long-term addition of animal manure does not affect the soil pore characteristics nor the soil hydraulic conductivity for the soil below ploughing depth.
3. A high level of nutrient application increases soil macroporosity and near-saturated hydraulic conductivity for the subsoil. This effect is independent of nutrient source.

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REFERENCES

- Anderson SH Gantzer CJ & Brown JR 1990. Soil physical properties after 100 years of continuous cultivation. *Journal of Soil and Water Conservation* 45, 117–121.
- Ankeny MD Ahmed M Kaspar TC & Horton R 1991. Simple field method for determining unsaturated hydraulic conductivity. *Soil Science Society of America Journal* 55, 467–470.
- Axelsen JA & Thorup-Kristensen K 2000. Collembola and mites in plots fertilised with different types of green manure. *Pedo Biologia* 44, 556–566.
- Ball BC & Schjønning P 2002. Air permeability. In: *Methods of soil analysis, Part 1*, eds JH Dane & GC Topp, 3rd edn. ASA & SSSA Madison WI pp 1141–1158.
- Ball BC Tebrügge F Sartori L Giráldez JV & González P 1998. Influence of no-tillage on physical, chemical and biological soil properties. In: *Experience with the applicability of no-tillage crop production in the West-European Countries*, eds F Tebrügge & A Böhrnsen, Final report, Concerted Action No. AIR 3-CT 93-1464 Fachverlag Köhler Giessen, Germany, pp 7–27.
- Bhatnagar VK Kundu S & Prakash VED 1992. Effect of long-term manuring and fertilization on soil-physical properties under soybean (*Glycine max*) – wheat (*Triticum aestivum*) cropping sequence. *Indian Journal of Agricultural Sciences* 62, 212–214.
- Blackwell PS Ringrose-Voase AJ Jayawardane NS Olsson KA McKenzie DC & Mason WK 1990. The use of air-filled porosity and intrinsic permeability to air to characterize structure of macropore space and saturated hydraulic conductivity of clay soils. *Journal of Soil Science* 41, 215–228.
- Campbell CA Schnitzer M Stewart JWB Biederbeck VO & Selles F 1986. Effect of manure and P fertilizer on properties of a black chernozem in southern Saskatchewan. *Canadian Journal of Soil Science* 66, 601–614.
- Celik I Ortas I & Kilic S 2004. Effects of compost, mycorrhiza, manure and fertilizer on some physical properties of a Chromoxerert soil. *Soil and Tillage Research* 78, 59–67.
- Chang AC Page AL & Varneke JE 1983. Soil conditioning effects of municipal sludge compost. *Journal of Environmental Engineering (ASCE)* 109, 574–583.
- Christensen BT 1997. The Askov long-term field experiments. *Archiv für Acker- und Pflanzenbau und Bodenkunde* 42, 265–278.
- Christensen BT Petersen J Kjellerup V & Trentemøller U 1994. The Askov long-term experiments on animal manure and mineral fertilizers: 1894–1994. Reports from the Danish Institute of Plant and Soil Science Foulum Report No. 43, 85 pp.
- Christensen O Daugbjerg P Hinge J Jensen JP & Sigurdardottir H 1987. The effects of cultivation practices on earthworms and their possible role as bioindicators. *Tidsskrift for Planteavl* 19, 15–32. [In Danish with English Summary]
- Comia RA Stenberg M Nelson P Rydberg T & Håkansson I 1994. Soil and crop responses to different tillage systems. *Soil and Tillage Research* 29, 335–355.
- Djurhuus J & Olesen JE 2000. Characterisation of four sites in Denmark for long-term experiments on crop rotations for organic farming. Reports from the Danish Institute of Agricultural Sciences Foulum Report No. 33 (Plant Production Series), 74 pp.
- Edmeades DC 2003. The long-term effects of manures and fertilisers on soil productivity and quality: a review. *Nutrient Cycling in Agroecosystems* 66, 165–180.
- Edwards WM Shipitalo MJ Traina SJ Edwards CA & Owens LB 1992. Role of *Lumbricus terrestris* (L.) burrows on quality of infiltrating water. *Soil Biology and Biochemistry* 24, 1555–1561.
- Franken H & Hurtmanns E 1985. Der Einfluss langjähriger Düngungsmaßnahmen auf die Dynamik der Bodenstruktur. *Zeitschrift für Pflanzenernährung und Bodenkunde* 148, 159–168.
- Gradwell MW 1961. A laboratory study of the diffusion of oxygen through pasture topsoils. *New Zealand Journal of Science* 4, 250–270.
- Groenevelt PH Kay BD & Grant CD 1984. Physical assessment of a soil with respect to rooting potential. *Geoderma* 34, 101–114.
- Goyal S Mishra MM Dhankar SS Kapoor KK & Batra R 1993. Microbial biomass turnover and enzyme activities following the application of farmyard manure to field soils with and without previous long-term applications. *Biology and Fertility of Soils* 15, 60–64.
- Hansen S 1996. Effects of manure treatment and soil compaction on plant production of a dairy farm system converting to organic farming practice. *Agriculture, Ecosystems and Environment* 56, 173–186.
- Håkansson I Grath T & Olsen HJ 1996. Influence of machinery traffic in Swedish farm fields on penetration resistance in the subsoil. *Swedish Journal for Agricultural Research* 26, 181–187.
- Iversen BV Koppelgaard M & Jacobsen OH 2004. An automated system for measuring air permeability and hydraulic conductivity in the laboratory on large soil cores. Reports from the Danish Institute of

- Agricultural Sciences Foulum Report No. 111 (Plant Production Series), 24 pp.
- Iversen BV Schjønning P Poulsen TG & Moldrup P 2001. In situ, on-site and laboratory measurements of soil air permeability: boundary conditions and measurement scale. *Soil Science* 166, 97–106.
- Jacobowitz LA & Steenhuis TS 1984. Compost impact on soil moisture and temperature. *Biocycle* 25, 56–60.
- Khaleel R Reddy KR & Overcash MR 1981. Changes in soil physical properties due to organic waste applications: a review. *Journal of Environmental Quality* 10, 133–141.
- Klute A & Dirksen C 1986. Hydraulic conductivity and diffusivity: Laboratory Methods. In: *Methods of soil analysis, Part 1*, 2nd edn. Physical and mineralogical methods, ed A Klute, ASA & SSSA Madison WI pp 687–734.
- Mathers AC & Stewart BA 1984. Manure effects on crop yields and soil properties. *Transactions of the American Society of Agricultural Engineers* 27, 1022–1026.
- Martens DA Johanson JB & Frankenberger Jr WT 1992. Production and persistence of soil enzymes with repeated additions of organic residues. *Soil Science* 153, 53–61.
- Metzger L & Yaron B 1987. Influence of sludge organic matter on soil physical properties. *Advances in Soil Science* 7, 141–163.
- Miller JJ Sweetland NJ & Chang C 2002. Hydrological properties of a clay loam soil after long-term cattle manure application. *Journal of Environmental Quality* 31, 989–996.
- Munkholm LJ 2000. The spade analysis – a modification of the qualitative spade diagnosis for scientific use. Reports from the Danish Institute of Agricultural Sciences Foulum Report No. 28 (Plant production Series), 73 pp.
- Munkholm LJ Schjønning P & Rasmussen KJ 2001a. Non-inversion tillage effects on soil mechanical properties of a humid sandy loam. *Soil and Tillage Research* 62, 1–14.
- Munkholm LJ Schjønning P & Petersen CT 2001b. Soil mechanical behaviour of sandy loams in a temperate climate: case-studies on long-term effects of fertilization and crop rotation. *Soil Use and Management* 17, 269–277.
- Munkholm LJ Schjønning P Deboz K Jensen JE & Christensen BT 2002b. Aggregate strength and mechanical behaviour of a sandy loam under long-term fertilization treatments. *European Journal of Soil Science* 53, 129–137.
- Nielsen JD & Møberg JP 1984. Classification of 5 soil profiles at Danish State Research Stations. *Tidsskrift for Planteavl* 88, 155–168. [In Danish with English summary].
- Petersen CT Hansen S Jensen HE Holm J & Bender Koch C 2004. Movement of suspended matter and a bromide tracer to field drains in tilled and untilled soil. *Soil Use and Management* 20, 271–280.
- SAS Institute 1996. SAS Institute SAS/STAT™ Software: Changes and Enhancements Through Release 6.11. SAS Institute Cary NC.
- Schjønning P 1985a. A laboratory method for determination of gas diffusion in soil. Reports from the Danish Institute of Plant and Soil Science Copenhagen Report No. S1773, 19 pp. [In Danish with English summary].
- Schjønning P 1985b. Soil pore characteristics. II. Effect of incorporation of straw and soil tillage. *Tidsskrift for Planteavl* 89, 425–433. [In Danish with English summary].
- Schjønning P 1986. Soil permeability by air and water as influenced by soil type and incorporation of straw *Tidsskrift for Planteavl* 90, 227–240. [In Danish with English summary].
- Schjønning P 1989. Long-term reduced cultivation II. Soil pore characteristics as shown by gas diffusivities and permeabilities and air-filled porosity. *Soil and Tillage Research* 15, 91–103.
- Schjønning P 1995. Measurements of soil physical properties in the Askov long-term experiments. In: *The Askov long-term experiments on animal manure and mineral fertilizers*, eds BT Christensen & U Trentemøller, Reports from the Danish Institute of Plant and Soil Science Foulum Report No. 29 pp 125–135.
- Schjønning P Christensen BT & Carstensen B 1994. Physical and chemical properties of a sandy loam receiving animal manure, mineral fertilizer or no fertilizer for 90 years. *European Journal of Soil Science* 45, 257–268.
- Schjønning P Elmholt S Munkholm LJ & Deboz K 2002a. Soil quality aspects of humid sandy loams as influenced by organic and conventional long-term management. *Agriculture, Ecosystems and Environment* 88, 195–214.
- Schjønning P Munkholm LJ Moldrup P & Jacobsen OH 2002b. Modelling soil pore characteristics from measurements of air exchange: the long-term effects of fertilization and crop rotation. *European Journal of Soil Science* 53, 331–339.
- Shirani H Hajabbasi MA Afyuni M & Hemmat A 2002. Effects of farm-yard manure and tillage systems on soil physical properties and corn yield in central Iran. *Soil and Tillage Research* 68, 101–108.
- Sundberg PS Callesen I Greve MH & Raulund-Rasmussen K 1999. Danish soil profiles. Report from the Danish Institute of Agricultural Sciences Tjele Denmark ISBN 87-88976-39-4. [In Danish]
- Taylor SA 1949. Oxygen diffusion as a measure of soil aeration. *Soil Science Society of America Proceedings* 14, 55–60.
- Tisdall JM & Oades JM 1982. Organic matter and water-stable aggregates in soils. *Journal of Soil Science* 33, 141–163.
- van den Elsen E Stolte J & Veerman G 1999. Three automated laboratory systems for determining the hydraulic properties of soils. In: *Characterization and measurement of the hydraulic properties of unsaturated porous media, Part 1*, eds MT van Genuchten FJ Leij & L Wu, US Salinity Laboratory, USDA, ARS. Department of Environmental Sciences University of California Riverside CA pp 329–340.
- Watts CW & Dexter AR 1998. Soil friability: theory, measurement and the effects of management and organic carbon content. *European Journal of Soil Science* 49, 73–84.
- Wei QF Lowery B & Peterson AE 1985. Effect of sludge application on physical properties of a silty clay loam soil. *Journal of Environmental Quality* 14, 178–180.
- Whalen JK Parmelee RW & Edwards CA 1998. Population dynamics of earthworm communities in corn agroecosystems receiving organic or inorganic fertilizer amendments. *Biology and Fertility of Soils* 27, 400–407.

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